

Gaussian Quantum Steering and Entanglement Dynamics in Coupled Lossy Waveguides



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Dedication

I dedicate this work to my parents, whose unconditional support and encouragement have been the backbone of my academic journey. I also dedicate it to my husband, whose confidence in me and constant support have greatly encouraged and strengthened my academic journey.

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Abstract

List of Abbreviations

CV	Continuous Variable
DV	Discrete Variable
1sDI	One-Sided Device-Independent
QKD	Quantum Key Distribution
1sDI-QKD	One-Sided Device-Independent Quantum Key Distribution
PPT	Positive Partial Transpose
TMGS	Two-Mode Gaussian State
NOON	N-Photon State
HOM	Hong-Ou-Mandel (interference)
QIT	Quantum Information Theory
EPR	Einstein-Podolsky-Rosen
EN	Entanglement via Logarithmic Negativity
E_N	Logarithmic Negativity Measure
$S^{A \rightarrow B}$	Steering from Party A to Party B
V	Covariance Matrix
ρ	Density Matrix
γ	Loss Rate (decay)
J	Coupling Strength (waveguides)
γ/J	Loss-to-Coupling Ratio

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Chapter 1

Introduction

1.1 Background and motivation

Quantum correlations is one of the basic aspect of quantum mechanics required for many current technologies in quantum information. The field of quantum information started with the aim of controlling individual quantum bits (qubits), and has led to important concepts, such as quantum computing, quantum cryptography, and quantum teleportations [5].

Later researchers discovered an immensely powerful alternative in using continuous-variable (CV) quantum information carriers instead of just qubits, offering the quantization of degrees of freedom of bosonic systems, such as the electromagnetic field. These systems are crucial in quantum communication, as well as to very advanced methods, which are only limited by quantum physics, used to detect, sense, and image [5].

In quantum processes involving CVs, it is frequently of interest to consider only the Gaussian realm, i.e., Gaussian states and Gaussian transformations, and this is a space where the maths is simple and well-organized, and where in the lab, the optical devices needed (such as beam splitters, phase shifters, and squeezers) are simple to realize and readily are available in vast quantities [5]. This area is referred to as Gaussian quantum information processing, and allows a vast range of important tasks to be performed. The most important of these protocols are quantum correlations, most notably entanglement and quantum steering, which is key to almost all quantum information protocols being implemented successfully [5].

In 1935 the idea of quantum entanglement gained much attention with the Einstein-Podolsky-Rosen (EPR) paradox raising the question of whether quantum mechanics provided a full description of reality. Later that year, Schrodinger referred to this effect as entanglement and also contributed to what is now called quantum steering. These discoveries formed the basis of the modern theory of quantum correlations [13, 19].

The phenomenon known as quantum steering, a form of quantum correlation, which has received considerable attention in quantum information theory, especially after its formalisation in 2007. Schrodinger had first described the phenomenon as magic as the action does not facilitate transmission of information, and it is the ability of one party (Alice) to measure or steer the wave function of the other, who is at a distance (Bob) [13]. One common notion of modern theory is the inability of the conditional quantum states of one party (Bob) to be described by a local hidden state (LHS) model. Steerable a

shared bipartite quantum state: In case the steerable shared bipartite quantum state of the two systems is described by the bipartite quantum state called ϱ_{AB} then Bob must be willing to accept the fact that Alice is acting upon his system by some scheme of action at a distance. On the other hand, when there is an existing LHS model, then the state of the system with variables is that of an unsteerable system, in which we refer to it as state $A B$ [13].

One-sided device-independent Quantum Key Distribution (QKD) is essential due to quantum steering, in which one party is confident in their measurement equipment, whereas the other is not, which is clear in the possibility of secret key distillation and demonstrates the state steerability. It also has randomness certification in this asymmetric case with more extractable randomness than the Bell case with full device-independence. Besides, steering offers an operational benefit in subchannel discrimination tasks and is conceptually equivalent to quantum teleportation and analysis of security in secret sharing protocols [13].

Quantum entanglement is recognized as a fundamental essence of quantum mechanics as well as the field of quantum information theory. Entanglement is described as a physical phenomenon, which is the property that the associated particles remain correlated irrespective of their physical distance between them. This property is deemed important to the operation of the quantum information processing systems. Quantum entanglement, found among the essences of quantum mechanics and quantum information theory, is deemed vital to quantum information processing. Its importance can be seen in numerous important technological domains with quantum computing, quantum cryptography, and quantum teleportation playing a significant part. Particularly, entanglement is viewed as being crucial to illustrate the supremacy of a quantum computer when comparing it to its classical analogs [27]. In coupled lossy waveguides, entanglement is an important concept, as these waveguides can be used as basic parts of quantum circuits. Although the real devices always have some loss, the devices are still useful to quantum information work and it is therefore necessary to know how entanglement behaves in them real [22].

An optical waveguide is a structure that guides light along a desired path. Coupled waveguides are very important because they act as basic building blocks for quantum circuits and allow efficient control of light and can be effectively used to manipulate light to accomplish quantum information processing, where the manipulation of individual photons in a waveguide is fundamentally involved. In addition, coupled silica

waveguides have been extended to make multimode integrated interferometers which can produce arbitrary quantum circuits, as well as can produce two and four photon entangled states analogous to NOON states. In addition to computation, such structures can be considered to study coherent phenomena such as random walks and other important quantum effects such as quantum interference, entanglement and quantum walk. An important point of the study is that it is dealing with real material loss and it still finds that the waveguides hold up well. They are able to maintain their entanglement even though the loss is quite strong. This shows that coupled lossy waveguides can still work as useful parts of quantum circuits, making them suitable for many quantum information applications [22].

As quantum technologies have advanced, optical waveguide-based photonic platforms have become the focus of modern quantum hardware. In these Waveguide devices light is trapped in small, stable cores with high index, and small, scalable and extremely scalable quantum circuits are constructed. By making two waveguides close to each other, all the evanescent fields overlaps, leading to evanescent coupling mechanism that enables controlled energy exchange just like a beam splitter. This interaction is the foundation on which to apply on-chip quantum gates and other necessary photonic operations are applied [22].

But, in real devices, some amount of optical loss and noise from the environment is always present. The energy of the photons decreases as they move, leading to an effective interaction with the empty vacuum modes; the interaction tends to destroy the quantum state of a system, and destroys the distinct quantum correlations between the modes. Such perturbations destroy quantum correlations, i.e quantum entanglement and quantum steering, and they pose an enormous limitation to photonic architectures which require either large propagation ranges or very precise interactions between gates [11, 22]

For this reason, understanding how quantum correlations behave in coupled waveguides when they experience losses is essential for determining whether integrated quantum photonic systems can work in practice. By studying how entanglement and steering change under attenuation, we can evaluate how robust these quantum resources are and assess the viability of waveguide-based circuits for future quantum technologies.

This research analyzes the time-dependent dynamics of quantum steering and entanglement in a passive two-mode directional coupler with realistic loss, using the Gaussian-

state formalism. The study examines how several key parameters—including the loss coefficient κ , the initial nonclassical depth τ , the purity μ , and the input squeezing r of the Gaussian state—collectively determine the preservation or degradation of quantum steering and entanglement as the fields propagate through lossy waveguides, while the coupling rate J is kept constant throughout the analysis.

1.2 Thesis layout

this thesis is structured as follows

chapter 2 begins with overview of continuous variable quantum states particularly gaussian states together with their representation through quadrature operators, wigner functions and covariance matrices. After that the discussion moves towards optical waveguides which highlights evanescent coupling, loss and decoherence in waveguides and applications of optical waveguides in quantum photonics. Moreover, the chapter then reviews the two important quantum correlations i.e quantum steering and quantum entanglement with their quantification, significance and applications. Last but not the least the applications and relevance of the optical waveguides in quantum circuits and photonics, and the applications of quantum steering and quantum entanglement are overviewed.

Chapter 2

Literature Review:

This chapter gives a review of the theoretical background of quantum correlations in integrated photonic circuits. It begins with a review of continuous-variable quantum states, particularly of Gaussian states, the properties of which are fully characterized by a covariance matrix, extracted out of the field quadratures. This is in turn followed by a discussion of the physical platform of optical waveguides, how the light is localized within the guiding medium and interaction between adjacent modes through evanescent interaction, provide ordinary optical attenuation (α) couples the system to the vacuum and causes the system to decay (Γ), making it hard to preserve quantum information. The chapter also summarizes two important quantum resources discussed in this work, quantum steering, denoted by the Gaussian steering measure, denoted by the symbol (S), and quantum entanglement, characterized by the logarithmic negativity E_N . All the above factors together are the conceptual framework that must be used to research the dynamical development of these resources in coupled waveguide architectures. This chapter ends with the discussion of the vital uses of entanglement, steering, and waveg-

uities.

2.1 Quantum states of light

2.1.1 Continuous variable states

In quantum mechanics, a Continuous variable state is a quantum state where the physical observables i.e momentum and position vary continuously unlike discrete variable state where observables can take only discrete values. These systems are illustrated with in an infinite-dimensional Hilbert space [1]. In these systems the principal variables are the quadrature operators which obey specific commutation relations that form the foundation of the system's quantum structure [4]. Continuous variable quantum states are classified into two categories: Gaussian and non Gaussian states [6].

2.1.2 Gaussian states

In a continuous variable system quantum system, a Gaussian state is a special kind of quantum state whose Fourier representation has a Gaussian form in phase space. These states are completely described by their first and second moments. The quadrature operators are

$$Q = \frac{1}{\sqrt{2}}(a + a^\dagger), \quad R = \frac{i}{\sqrt{2}}(a^\dagger - a), \quad (2.1)$$

which satisfy the canonical commutation relation

$$[Q, R] = i. \quad [4] \quad (2.2)$$

The Wigner function of a Gaussian state is

$$W(\mathbf{z}) = \frac{1}{(2\pi)^m \sqrt{\det \Sigma}} \exp \left[-\frac{1}{2}(\mathbf{z} - \bar{\mathbf{z}})^T \Sigma^{-1}(\mathbf{z} - \bar{\mathbf{z}}) \right]. \quad (2.3)$$

Here,

$$\mathbf{z} = (Q_1, R_1, \dots, Q_m, R_m)^T$$

is the phase-space vector,

$$\bar{\mathbf{z}} = \langle \hat{\mathbf{z}} \rangle$$

is the displacement vector, and

$$\Sigma_{ij} = \frac{1}{2} \langle \hat{z}_i \hat{z}_j + \hat{z}_j \hat{z}_i \rangle - \langle \hat{z}_i \rangle \langle \hat{z}_j \rangle$$

is the covariance matrix.

All physical properties of the gaussian states such as purity, entropy, steering and entanglement can be determined directly from the covariance matrix without full density operator because of the gaussian form of weigner function. typical examples of gaussian states include the coherent states, squeezed states, thermal states and vaccume state [5, 6].

2.2 Covariance matrix

In continuous-variable quantum systems, the degrees of freedom of the bosonic modes are represented by the quadrature operators $\hat{Q}_k$ and $\hat{P}_k$. The phase-space vector $\hat{\mathbf{z}}$ is defined as

$$\hat{\mathbf{z}} = (\hat{Q}_1, \hat{P}_1, \dots, \hat{Q}_m, \hat{P}_m)^T. \quad (2.4)$$

The covariance matrix Σ for an n -mode quantum state is defined as

$$\Sigma_{ij} = \frac{1}{2} \langle \hat{z}_i \hat{z}_j + \hat{z}_j \hat{z}_i \rangle - \langle \hat{z}_i \rangle \langle \hat{z}_j \rangle. \quad (2.5)$$

Here, Σ is a real, symmetric matrix that must satisfy the quantum uncertainty principle in order to represent a physical Gaussian state [7].

2.3 Optical waveguides as Quantum systems

An Optical waveguide is a structure that guide light along a desired path. It consists of thin, high refractive index core layer positioned between low refractive index cladding materials and the light is trapped within the core through total internal refrelction. Photons passes through these waveguides in particular patterns known as modes [8].

2.3.1 Evanscent overlap/coupled mode theory

When light passes through a waveguide system, the electric feild is mostly trapped in the high refractive index core. Yet, a part of the feild can penetrate into the surround-

ing lower-index cladding. We call this decaying field as the evanescent field. If we place another waveguide in this region i.e. the evanescent field reach typically less than 1-2 wavelengths away light can tunnel between these two waveguides. We call this transfer of energy through coupling of evanescent fields as evanescent coupling. The coupling strength between the waveguides depends on the separation distance between the waveguides, refractive index and mode shapes [9].

2.3.2 loss and decoherence in real waveguide systems

In real optical waveguide system, the performance is undermined by two processes: classical optical power loss (attenuation α) and quantum information loss (decoherence, Γ). Exponential decay of optical intensity as light propagates is called attenuation while decoherence is the process in which a quantum system loses its correlation properties i.e. superposition or entanglement due to interaction with the environment, results in losing the necessary phase relationship and limiting visibility of interference [10]. In quantum photonics the two problems are interconnected, classical optical loss (α) act as primary source of quantum decoherence. When a photon is lost from the waveguide system, its quantum state becomes coupled to unobserved vacuum port which results in transferring quantum information to the environment irreversibly and thus decoheres the entanglement and other nonclassical quantum resources. This loss mechanism is as an amplitude damping channel [11].

2.3.3 Relevance of optical waveguides for quantum circuits/photonics

The optical waveguides are highly necessary in quantum photonics since they have facilitated the researchers to replace large laboratory systems with small chips that can soon be used in quantum technologies. All structures are built on the chip and this enables the researchers to implement quantum logic gates like CNOT, Toffoli and Fredkin gate on a chip. They find application in building quantum computers, secure communication and quantum sensors. Low loss photonic materials like silicon, silicon nitride and lithium niobate are highly essential in this fact since optical waveguides are very important in the character of quantum information processing of integrated photonic circuits. Their nonlinear and optical properties can be optimally used in control of generation, manipulation, and detection of quantum states with very low propagation loss. When photon sources, beam splitters, phase shifters, and detectors are implemented on a single chip on a waveguide basis, the resultant [11, 12].

2.4 Steering in continuous variable

Quantum steering is a quantum correlation that was first discovered in the 1935 argument by Einstein, Podolsky, and Rosen (EPR) on the completeness of quantum mechanics. Erwin Schrodinger noted that a party (Alice) could control, or "direct the wave function of a remote party (Bob) simply by measuring a local system in a manner that was appropriate, which he called as "magic" [13]. In modern quantum information theory, quantum steering is operationally characterized by the fact that it is not possible to describe the set of conditional quantum states that Bob (the steering assemblage) measures in terms of a Local Hidden State (LHS) model [13, 14]. A steering model assumes that the quantum state of Bob has a hidden state, and Alice only provides a measurement of the distribution of that hidden state, which is steerable as well as genuine entangled. Steering occupies a definite middle reference between entanglement and Bell nonlocality, that is, steerable times quantum correlations can be established, and entangled times quantum correlations can be verified in a one-sided device-independent scenario, where it is assumed that the measurement apparatus of Alice is never trusted (characterized) [13, 14]. In continuous-variable (CV) systems generally implemented with optical fields (quadratures X and P), the steering-detection problem is often modeled in terms of criteria based upon the Heisenberg uncertainty relation of conditional measurements. The steering criterion of a state shared between A and B is the condition that the product of the minimum inferred variances of the position of Bob, in quadratures of the momentum (X_B) and momentum (P_B) quadratures, in terms of the measurements of Alice, be lower than the quantum limit: [15, 13]

$$V_{\text{inf}}(X_B) V_{\text{inf}}(P_B) < 1. \quad (2.6)$$

For CV Gaussian states where both the states and measurements are Gaussian, this EPR criterion is both necessary and sufficient to detect bipartite steering [15]. Remarkably, a bipartite Gaussian state is unsteerable with Gaussian measurements if and only if a specific matrix inequality involving the covariance matrix V_{AB} is satisfied:

$$V_{AB} + i(0_A \oplus \Omega_B) \geq 0. \quad (2.7)$$

If the state is steerable with Gaussian measurements, it is also steerable with a certain pair of canonical quadratures [13].

2.5 Quantification of Quantum Steering

2.5.1 Gaussian Steering Measure

The Gaussian $A \rightarrow B$ steerability $\mathcal{S}_{A \rightarrow B}(\sigma_{AB})$ is defined based on the symplectic eigenvalues $\{\bar{\nu}_j^B\}$ of the Schur complement matrix M_{σ_B} , calculated from the state's covariance matrix σ_{AB} :

$$\mathcal{S}_{A \rightarrow B}(\sigma_{AB}) := \max \left\{ 0, - \sum_{\bar{\nu}_j^B < 1} \ln(\bar{\nu}_j^B) \right\}. \quad (2.8)$$

For the specific case of two-mode Gaussian states (where the steered party has only one mode), the measure simplifies to the Rényi-2 coherent information:

$$\mathcal{S}_{A \rightarrow B}(\sigma_{AB}) = \max \left\{ 0, \frac{1}{2} \ln \left(\frac{\det \sigma_A}{\det \sigma_{AB}} \right) \right\} = \max \{0, H_2(\sigma_A) - H_2(\sigma_{AB})\}, \quad (2.9)$$

where $H_2(\sigma)$ denotes the Rényi-2 entropy.

Properties and Hierarchy: The Gaussian steering measure satisfies several desirable mathematical properties: it is invariant under local unitaries, convex, additive, and monotonically non-increasing under quantum operations performed on the untrusted party (Alice). It vanishes if and only if the state is nonsteerable by Gaussian measurements. Moreover, it respects the hierarchy of correlations:

$$\mathcal{S}_{A \rightarrow B}(\sigma_{AB}) \leq \mathcal{E}_2(\sigma_{AB}), \quad (2.10)$$

where $\mathcal{E}_2$ denotes the Gaussian Rényi-2 measure of entanglement. For pure states,

$$\mathcal{S}_{A \rightarrow B}(\sigma_{AB}^p) = \mathcal{E}_2(\sigma_{AB}^p). \quad (2.11)$$

Operational Meaning: For two-mode Gaussian states, Gaussian steerability has a direct operational interpretation in one-sided device-independent QKD:

$$\mathcal{K} \geq \max \{0, \mathcal{S}_{B \rightarrow A}(\sigma_{AB}) + \ln 2 - 1\}, \quad (2.12)$$

where $\mathcal{K}$ is the guaranteed secure key rate[16].

2.6 Entanglement in continuous variable systems

Entanglement is essentially known to be an inherent property of quantum theory, and it is the main resource of most quantum information science and technologies [17]. Originating from conceptual structures identified by Einstein, Podolsky, and Rosen (EPR) and Schrödinger, entanglement describes nonlocal correlations within composite quantum systems that cannot be replicated by classical systems [19]. It is a physical fact that two or more particles show correlations which cannot be explained classically, and these particles will be correlated no matter how far they are from each other [27]. Historically considered a "spooky" phenomenon, entanglement is now made part of engineering essential quantum protocols such as quantum computing and quantum teleportation in CV systems associated with infinite dimensional Hilbert spaces, provide the advantage of allowing deterministic and unconditional quantum operations of which is often a shortcoming in DV schemes due to relying on imperfect single-photon generation and detection [17]. Within the CV framework, the family of Gaussian states is the most applicable one, and it is the subject of entanglement research [20]. The quantitative effort to determine the intensity or the extent of entanglement within a quantum system is termed entanglement quantification. Quantification is needed because the measured degree of entanglement directly influences the performances of quantum information tasks [27].

2.6.1 Entanglement Quantification Measures

To measure the entanglement created and maintained in a system in a quantitative way, scientists use the special measures. The major measure in the reviewed studies is the logarithmic negativity E_N .

E_N is defined as

$$E_N(t) = \log_2 \|\rho^T\|, \quad (2.13)$$

where ρ^T denotes the partial transpose of the density matrix ρ , and $\|\cdot\|$ represents the trace norm. A non-zero value for E_N confirms that the state is entangled. For Gaussian states, such as squeezed states, the measure of entanglement, E_N , is determined by the symplectic eigenvalues ($\tilde{\nu}$) of the covariance matrix (σ). The state is entangled if the smallest symplectic eigenvalue, $\tilde{\nu}_<$, satisfies $\tilde{\nu}_< < \frac{1}{2}$. Logarithmic negativity in this context is given by

$$E_N(t) = \max[0, -\ln(2\tilde{\nu}_<(t))], \quad (2.14)$$

and serves as an upper bound to the distillable entanglement[22].

2.7 loss and decoherence

Decoherence has become a prominent subject in modern physics, It is propelled by the desire to know the border to the classical world and quantum, and the manner in which it is detrimental to the processing of quantum information [23].Fundamentally,decoherence is nothing but the irreversible, uncontrollable, and irreversible accumulation of quantum correlations, such as entanglement, between a system and its environment [23, 24]. Decoherence has attracted a lot of interest because of its relevance in explaining the quantum-classical boundary as well as its harmful effects on quantum information processing technologies.The main mechanism is that the environment actively monitors some system observables and hence, the decoherence of non- preferred states takes place. This monitoring thus imposes the choice of pointer states, which are robust and stable to the encounter of the environment. These pointer states continue to correlate with the surrounding universe in spite of being immersed in the surrounding environment. The ensuing fast conversion of quantum superpositions to mixtures is the reason why so simple superposition states, which are commonly known as "Schrodinger cat" states, do not seem to appear.The selection of these preferred states is termed Environment-Induced Superselection (Einselection).Einselection places a basic limitation on the bulk of the system Hilbert space, causing it to be restricted to a small classical domain that is made up of pointer states [25].

2.7.1 Loss (Dissipation) and Pure Decoherence

Decoherence is usually accompanied by dissipation, which refers to the net exchange of energy with the environment [23].But there are certain theoretical cases that focus on pure decoherence (or dephasing) that is, there is no exchange of energy.The phenomenon of pure decoherence is of special interest in explaining why macroscopic superpositions do not occur, since the mechanism that leads to this is to be proposed to rely on the distinguishability of the states, and not on the difference in their energy [23].Pure decoherence can mostly happen either when the system Hamiltonian (H_S) is systematically approximately commuting with the system bath interaction Hamiltonian (H_{int}),or when the system initial states vary at a rate significantly slower than the decoherence timescale. Pure decoherence is considered the most important in theoretical models

which attempt to explain the apparent lack of superpositions of macroscopically distinguishable states, including Schrodinger-cat states [23]. The functions of these phenomena should be differentiated: relaxation and noise (loss) are the situations in which the environment perturbs the system, decoherence and einselection are the situations in which the system perturbs the environment [25]. Another extremely significant thing about decoherence, particularly in macroscopic systems, is the separation of time scales between decoherence and energy loss (relaxation).the rate at which a system decoheres is oftentimes dependent on the distance between the superposed robust states of states that are macroscopically distinct, the dimensionless distance D can be very large (e.g., 10^{20}), so decoherence times can be very fast, even when the rate of energy relaxation of the system is very small [26].

2.7.2 Loss mechanism in waveguides

Loss (κ) in waveguides is due to loss in the material of the waveguide structure itself: Waveguides have often been designed, for example, to be identical, and so the loss rate for the field modes ($a1$ and $a2$) is usually taken to be the same. Loss is theoretically modelled in the framework of system–reservoir interaction by using a quantum–Liouville equation of motion. This is the equation of evolution of the density operator (ρ) with two terms, the Hamiltonian term to the unitary coupling and the loss term ($L\rho$) to represent the dissipation:

$$\dot{\rho} = -\frac{i}{\hbar}[H, \rho] + L\rho, \quad (2.15)$$

where the loss operator $L\rho$ specifically represents the decay rates of the modes $a1$ and $a2$ [22].

2.8 Applications and motivations

2.8.1 Applications of Quantum Entanglement

The concept of quantum entanglement is a pillar of quantum information theory and quantum mechanics and is used to explain highly correlated particles that become linked regardless of the distance separating them [27]. This has a critical role in proving the theoretical and algorithmic superiority of a quantum computers over its classical counterparts [27, 28]. An important element of quantum computing is the manipulation of entanglement that results in the enhancement of capabilities [28], and is applied in fun-

damental quantum information processing systems like quantum cryptography, quantum teleportation and quantum computing [27]. The quantification of entanglement is important in the determination of the intactness and the depth of the level of correlation that has a direct impact on the execution of quantum information tasks. Quantified entanglement measures include concurrence, negativity (Including logarithmic negativity), Schmidt decomposition, and entanglement of formation [27]. In case of bipartite Gaussian states, entanglement can be quantitatively measured by the logarithmic negativity of input states, such as squeezed states and photon number states, and entanglement dynamics is quantitatively measured in terms of logarithmic negativity. In waveguide architecture, the sustainability of generated entanglement can be significantly important in practical implementation of quantum communication and computation tasks, and dynamics of entanglement is studied by entanglement logarithmic negativity of inputs states [22]. Highly entangled states are essential in doing complex quantum information processing and implementing error-correcting code [28].

2.8.2 Applications of Quantum Steering

Quantum steering (QS) is a form of quantum correlation that has been formalized by Schrodinger which allows one party, Alice, to change the state of a remote party, Bob, remotely, by acting on local measurements which make use of common entanglement [28]. One of the most practical uses of quantum steering is the realization of desired quantum states, the so-called Measurement-Induced Quantum Steering (MIQS), a protocol that can be used to prepare a target quantum state regardless of the initial (mixed) state, in other words, the preparation of quantum states [28]. Quantum Steering plays an important role in offering security to one-sided device-independent Quantum Key Distribution (1SDI QKD) protocols, where the amount of steerability (Gaussian) can be used to measure the rate of guaranteed key rate [16]. Quantum steering also has a close connection with quantum teleportation, any gaussian state that can perform safe teleportation must be two way steerable [29]. Quantum steering also plays an important role in quantum-enhanced metrology [30].

2.8.3 Applications of Waveguides

Coupled waveguide designs are significant fundamental components that scientists examine whether they can be employed as straightforward building blocks to quantum circuits in integrated photonic devices. These are discrete optical systems that are quite

efficient in controlling the movement of light. Waveguides are useful in executing key activities in quantum computing. They, for instance, allowed a CNOT gate in a single silicon chip that consists of connected silica waveguides. Multimode interferometers are also easily constructed using waveguides. These are able to fabricate any quantum circuit and are an important component in the development of the quantum circuit behind the Shor's Quantum Factoring Algorithm on a photonic chip. Outside of the field of computing, waveguide arrays are used to investigate coherent phenomena such as quantum walks and a discrete form of the Talbot effect [22].

Chapter 3

Model System and Theoretical Formulations

3.1 Physical Model and Assumptions

3.1.1 Physical System Configuration

our system consist of two identical single mode waveguides which are placed parallel to each other in a close distance such that their evanescent feild overlap and enable coupling between them. The optical modes in each waveguide are represented by creation and anihilation operator a_1 and a_2 where a_1 represents optical mode in waveguide a_1 and a_2 represents optical mode in waveguide a_2 . Each waveguide mode act as a bosonic harmonic oscillator. The two waveguides are assumed to identical and at the same resonance frequency, and loss rate, which is a realistic assumption.

3.2 Hamiltonian Description of the Coupled Waveguide System

3.2.1 System Hamiltonian

The hamiltonian for our system is given by

$$H = \hbar\omega \left(a_1^\dagger a_1 + a_2^\dagger a_2 \right) + \hbar J \left(a_1^\dagger a_2 + a_2^\dagger a_1 \right). \quad (3.1)$$

Here, a_1 and a_2 are the bosonic field operators for the optical modes in each waveguide, satisfying the commutation relations $[a_1, a_1^\dagger] = 1$ and $[a_2, a_2^\dagger] = 1$.

the symbol ω is angular frequency of the each waveguide mode and the parameter J in the second part of the hamiltonian represents the coupling strength between the two waveguides and its value depends on the seperation distance between the waveguides.

The hamiltonian consists of two main parts:

Free energy terms: correspond to the free energy of the individual waveguide modes,

$$\hbar\omega \left(a_1^\dagger a_1 + a_2^\dagger a_2 \right).$$

Coupling terms: account for the evanescent coupling between the two waveguide modes,

$$\hbar J \left(a_1^\dagger a_2 + a_2^\dagger a_1 \right).$$

This term is responsible for the transfer of energy between the two waveguide modes and is the main focus of interest in this work.

3.2.2 Physical Interpretation of the Coupling

The evanescent interaction between the two parallel optical waveguides is taken into account by the coupling term in the system Hamiltonian. With the waveguides laid near enough to one another, the electromagnetic field about any given guided mode extends past the center of the waveguide and some of it is overlapped by a neighboring waveguide. It is this overlap that allows the coherent exchange of photons between the two modes.

The amplitude of the coupling between the waveguides is defined by the parameter J which defines the speed with which energy moves between the waveguides. The higher is the value of J the higher is the mode overlap and the speed of oscillatory exchanging of photons when $J=0$ two entirely independent and uncoupled waveguides are considered. Physically, such an interaction is similar to a beam splitter with the photons coherently able to be simultaneously hopped between the two modes without loss.

Without the dissipation, the coupling results in a periodic oscillation of photon population in the two waveguides. In the case of losses, such oscillations are slowly damped, although the coupling term still plays the role of creating and maintaining quantum correlations like coherence, entanglement and steering of the two modes. The coupling is therefore up to the center of the relevance of the allocation and maintenance of quantum interactions in integrated photonic systems.

3.3 Open-System Dynamics and Loss Modelling

The optical modes in realistic integrated photonic systems necessarily experience the effects of. absorbed and scattered gains and imprecise confinement in the. waveguides. Because of this, the coupled waveguide system cannot be implied as an isolated. system is a quantum system and it needs to be formulated in the context of an open quantum. systems. To include these dissipative effects, the system dynamics is given by using a density-matrix functional, in which the interaction with the external environment results. to decoherence and to energy decay.

The time development of the coupled waveguide system is contrived in this section in terms of. the Lindblad form of the quantum Liouville (master) equation. This approach allows for a regular addition of photon loss without introducing a change of state nature and maintain, the nature of the states, namely Gaussian. that is necessary to the further analysis of quantum steering and entanglement.

3.3.1 Quantum Liouville (Master) Equation

we can determine how this system evolve with loss by quantum liouville equation.

$$\dot{\rho} = -\frac{i}{\hbar}[H, \rho] + \mathcal{L}\rho \quad (3.2)$$

this equation is know as quantum liouville equation.it has two parts:

The Commutator:

$$[H, \rho]$$

which generates time evolution and is responsible for oscillatory exchange of energy and quantum correlation between the waveguides due to J.

Lindblad($\mathcal{L}\rho$) term: deals with material loss, decoherence and decays due to interaction with the environment

3.3.2 Lindblad Operators for Waveguide Loss

$$\begin{aligned} \mathcal{L}\rho = & -k_1 \left(a_1^\dagger a_1 \rho + \rho a_1^\dagger a_1 - 2a_1 \rho a_1^\dagger \right) \\ & - k_2 \left(a_2^\dagger a_2 \rho + \rho a_2^\dagger a_2 - 2a_2 \rho a_2^\dagger \right), \end{aligned} \quad (3.3)$$

k_1 and k_2 represents the loss rates of modes a and b , and $a, a^\dagger$ are the annihilation and creation operators for mode a , while ρ is the density matrix. b and $b^\dagger$ are vice versa for mode b .

3.4 Dynamical Equations of the Model System

This section is dealing with the derivation of the dynamical equations of the coupled. waveguide system. Through the calculation of the level of expectation values, the open-system dynamics may be defined as a closed set of second-order moments, which is

sufficient to describe Gaussian states.

3.4.1 Expectation-Value Formalism

The time evolution of any system observable O is obtained from

$$\frac{d}{dt}\langle O \rangle = \text{Tr}(O\dot{\rho}) = -i\langle [O, H] \rangle + \langle \mathcal{L}^\dagger(O) \rangle. \quad (3.4)$$

3.4.2 Definition of Relevant Second-Order Moments

$$n_1(t) = \langle a_1^\dagger(t) a_1(t) \rangle, \quad (3.5)$$

$$n_2(t) = \langle a_2^\dagger(t) a_2(t) \rangle, \quad (3.6)$$

$$m_1(t) = \langle a_1(t) a_1(t) \rangle, \quad (3.7)$$

$$m_2(t) = \langle a_2(t) a_2(t) \rangle, \quad (3.8)$$

$$c(t) = \langle a_1^\dagger(t) a_2(t) \rangle, \quad (3.9)$$

$$s(t) = \langle a_1(t) a_2(t) \rangle. \quad (3.10)$$

Here, $n_1(t)$ and $n_2(t)$ represent the mean photon numbers in the first and second waveguides, respectively. The moments $m_1(t)$ and $m_2(t)$ describe single-mode squeezing correlations, while $c(t)$ and $s(t)$ quantify inter-mode correlations responsible for quantum entanglement and steering between the two waveguides. o be written)

3.5 Moment Equations of Motion

3.5.1 Coupled Differential Equations

Using $\frac{d}{dt}\langle O \rangle = \text{Tr}(O\dot{\rho})$, the equations of motion for the relevant second-order moments are obtained as

$$\frac{d}{dt}\langle a_1^\dagger a_1 \rangle = iJ\left(\langle a_2^\dagger a_1 \rangle - \langle a_1^\dagger a_2 \rangle\right) - k_1\langle a_1^\dagger a_1 \rangle, \quad (3.11)$$

$$\frac{d}{dt}\langle a_2^\dagger a_2 \rangle = iJ\left(\langle a_1^\dagger a_2 \rangle - \langle a_2^\dagger a_1 \rangle\right) - k_2\langle a_2^\dagger a_2 \rangle, \quad (3.12)$$

$$\frac{d}{dt}\langle a_1^\dagger a_2 \rangle = iJ\left(\langle a_2^\dagger a_2 \rangle - \langle a_1^\dagger a_1 \rangle\right) - \frac{k_1 + k_2}{2}\langle a_1^\dagger a_2 \rangle, \quad (3.13)$$

$$\frac{d}{dt}\langle a_2^\dagger a_1 \rangle = iJ\left(\langle a_1^\dagger a_1 \rangle - \langle a_2^\dagger a_2 \rangle\right) - \frac{k_1 + k_2}{2}\langle a_2^\dagger a_1 \rangle, \quad (3.14)$$

$$\frac{d}{dt}\langle a_1 a_1 \rangle = -2iJ\langle a_1 a_2 \rangle - k_1\langle a_1 a_1 \rangle, \quad (3.15)$$

$$\frac{d}{dt}\langle a_2 a_2 \rangle = -2iJ\langle a_1 a_2 \rangle - k_2\langle a_2 a_2 \rangle, \quad (3.16)$$

$$\frac{d}{dt}\langle a_1 a_2 \rangle = -iJ\left(\langle a_1 a_1 \rangle + \langle a_2 a_2 \rangle\right) - \frac{k_1 + k_2}{2}\langle a_1 a_2 \rangle. \quad (3.17)$$

These equations form a closed set due to the quadratic form of the Hamiltonian and the Gaussian-preserving nature of the Lindblad dissipation.

3.5.2 Reduction of Independent Variables

Although seven coupled moment equations are obtained, not all of them are independent. The inter-mode coherence moments satisfy the Hermitian relation

$$\langle a_2^\dagger a_1 \rangle = \langle a_1^\dagger a_2 \rangle^*. \quad (3.18)$$

As a result, the system dynamics can be fully described by six independent second-order moments, while the remaining coherence follows directly from this relation.

3.6 Analytical Solutions of the Moment Equations

3.6.1 Equal-Loss Approximation

For identical waveguides, we assume equal loss rates,

$$k_1 = k_2 = \kappa. \quad (3.19)$$

3.6.2 General Analytical Solutions

Without imposing any restriction on the initial correlations, the exact solutions of the moment equations are given by

$$\begin{aligned}\langle a_1^\dagger a_1 \rangle(t) = & \frac{e^{-\kappa t}}{2} \left[\langle a_1^\dagger a_1 \rangle_0 + \langle a_2^\dagger a_2 \rangle_0 + (\langle a_1^\dagger a_1 \rangle_0 - \langle a_2^\dagger a_2 \rangle_0) \cos(2Jt) \right. \\ & \left. + i(\langle a_2^\dagger a_1 \rangle_0 - \langle a_1^\dagger a_2 \rangle_0) \sin(2Jt) \right],\end{aligned}\quad (3.20)$$

$$\begin{aligned}\langle a_2^\dagger a_2 \rangle(t) = & \frac{e^{-\kappa t}}{2} \left[\langle a_1^\dagger a_1 \rangle_0 + \langle a_2^\dagger a_2 \rangle_0 - (\langle a_1^\dagger a_1 \rangle_0 - \langle a_2^\dagger a_2 \rangle_0) \cos(2Jt) \right. \\ & \left. - i(\langle a_2^\dagger a_1 \rangle_0 - \langle a_1^\dagger a_2 \rangle_0) \sin(2Jt) \right],\end{aligned}\quad (3.21)$$

$$\begin{aligned}\langle a_1^\dagger a_2 \rangle(t) = & e^{-\kappa t} \left[\frac{\langle a_1^\dagger a_2 \rangle_0 + \langle a_2^\dagger a_1 \rangle_0}{2} + \frac{\langle a_1^\dagger a_2 \rangle_0 - \langle a_2^\dagger a_1 \rangle_0}{2} \cos(2Jt) \right. \\ & \left. - \frac{i}{2} (\langle a_1^\dagger a_1 \rangle_0 - \langle a_2^\dagger a_2 \rangle_0) \sin(2Jt) \right],\end{aligned}\quad (3.22)$$

$$\begin{aligned}\langle a_2^\dagger a_1 \rangle(t) = & e^{-\kappa t} \left[\frac{\langle a_1^\dagger a_2 \rangle_0 + \langle a_2^\dagger a_1 \rangle_0}{2} - \frac{\langle a_1^\dagger a_2 \rangle_0 - \langle a_2^\dagger a_1 \rangle_0}{2} \cos(2Jt) \right. \\ & \left. + \frac{i}{2} (\langle a_1^\dagger a_1 \rangle_0 - \langle a_2^\dagger a_2 \rangle_0) \sin(2Jt) \right],\end{aligned}\quad (3.23)$$

$$\begin{aligned}\langle a_1 a_1 \rangle(t) = & \frac{e^{-\kappa t}}{2} \left[\langle a_1 a_1 \rangle_0 - \langle a_2 a_2 \rangle_0 + (\langle a_1 a_1 \rangle_0 + \langle a_2 a_2 \rangle_0) \cos(2Jt) - 2i \langle a_1 a_2 \rangle_0 \sin(2Jt) \right],\end{aligned}\quad (3.24)$$

$$\begin{aligned}\langle a_2 a_2 \rangle(t) = & \frac{e^{-\kappa t}}{2} \left[-(\langle a_1 a_1 \rangle_0 - \langle a_2 a_2 \rangle_0) + (\langle a_1 a_1 \rangle_0 + \langle a_2 a_2 \rangle_0) \cos(2Jt) - 2i \langle a_1 a_2 \rangle_0 \sin(2Jt) \right],\end{aligned}\quad (3.25)$$

$$\langle a_1 a_2 \rangle(t) = e^{-\kappa t} \left[\langle a_1 a_2 \rangle_0 \cos(2Jt) - \frac{i}{2} (\langle a_1 a_1 \rangle_0 + \langle a_2 a_2 \rangle_0) \sin(2Jt) \right].\quad (3.26)$$

3.6.3 Special Case: Independently Prepared Input Modes

For independently prepared input modes, the initial inter-mode correlations vanish,

$$\langle a_1^\dagger a_2 \rangle_0 = \langle a_1 a_2 \rangle_0 = 0,$$

leading to simplified expressions that are used throughout the remainder of this thesis.

3.6.4 Compact Representation Using Gaussian Moments

For convenience, we give the following names to the moments:

$$n_1(t) = \langle a_1^\dagger a_1 \rangle(t), \quad n_2(t) = \langle a_2^\dagger a_2 \rangle(t),$$

$$c(t) = \langle a_1^\dagger a_2 \rangle(t), \quad m_1(t) = \langle a_1 a_1 \rangle(t), \quad m_2(t) = \langle a_2 a_2 \rangle(t), \quad s(t) = \langle a_1 a_2 \rangle(t).$$

Defining the constants

$$C_1 = \langle a_1^\dagger a_1 \rangle_0 + \frac{1}{2}, \quad C_2 = \langle a_2^\dagger a_2 \rangle_0 + \frac{1}{2},$$

$$D_1 = \langle a_1 a_1 \rangle_0, \quad D_2 = \langle a_2 a_2 \rangle_0,$$

The term $\frac{1}{2}$ accounts for the unavoidable vacuum (zero-point) fluctuations of the bosonic field.

The Gaussian moments evolve as:

$$\begin{aligned} n_1(t) &= \frac{e^{-\kappa t}}{2} [(C_1 + C_2 - 1) + (C_1 - C_2) \cos(2Jt)], \\ n_2(t) &= \frac{e^{-\kappa t}}{2} [(C_1 + C_2 - 1) - (C_1 - C_2) \cos(2Jt)], \\ c(t) &= \frac{i e^{-\kappa t}}{2} (C_2 - C_1) \sin(2Jt), \\ m_1(t) &= \frac{e^{-\kappa t}}{2} [(D_1 + D_2) \cos(2Jt) + (D_1 - D_2)], \\ m_2(t) &= \frac{e^{-\kappa t}}{2} [(D_1 + D_2) \cos(2Jt) + (D_2 - D_1)], \\ s(t) &= -\frac{i e^{-\kappa t}}{2} (D_1 + D_2) \sin(2Jt). \end{aligned}$$

These expressions are used directly in the numerical evaluations of Gaussian steering and entanglement throughout this thesis.

A single-mode Gaussian state is fully characterized by the parameters of its covariance matrix. The diagonal element c_j quantifies the local quadrature variance of the j -th mode and is directly related to the mean photon number according to

$$\langle a_j^\dagger a_j \rangle_0 = c_j - \frac{1}{2}, \tag{3.27}$$

where the term $\frac{1}{2}$ accounts for the unavoidable vacuum (zero-point) fluctuations of the bosonic field.

Within the nonclassicality–purity parametrization, the same quantity c_j can be expressed in terms of the nonclassicality τ_j and the purity μ_j as

$$c_j = \frac{\sqrt{\tau_j^2 - \tau_j + \frac{1}{(2\mu_j)^2} - \frac{1}{4}}}{1 - 2\tau_j} + \frac{1}{2}, \quad (3.28)$$

which guarantees that the corresponding covariance matrix satisfies the Heisenberg uncertainty principle.

The off-diagonal covariance parameter d_j is defined through the anomalous second-order moment

$$\langle a_j a_j \rangle_0 = -d_j, \quad (3.29)$$

where $d_j = |d_j|e^{i\phi_j}$ is, in general, a complex quantity. Within the same nonclassicality–purity parametrization, its magnitude is given by

$$|d_j| = \frac{\sqrt{\tau_j - \tau_j^2 + \frac{1}{(2\mu_j)^2} - \frac{1}{4}}}{1 - 2\tau_j}. \quad (3.30)$$

These two parameters, c_j and d_j , together fully specify the single-mode Gaussian input state, with c_j determining the phase-insensitive fluctuations and d_j encoding the phase-sensitive correlations of the field [31].

3.7 Covariance Matrix and Quantification of Quantum Steering and Entanglement

Moments obtained in this chapter are entirely sufficient to determine the quantum state of the system at all times.

The moments, $n_1(t)$, $n_2(t)$, $m_1(t)$, $m_2(t)$, $c(t)$, $s(t)$ computes the covariance of the optical field of two modes. This covariance matrix is the basic object of measuring the Gaussian quantum correlations. These moments are directly used to calculate using the covariance matrix formalism, the quantification of quantum steering and quantum entanglement. The analytic expressions derived in this chapter creates a connection be-

tween the physical parameters(strength of coupling, loss rate and initial state properties) of the waveguide system and the dynamical behavior of quantum correlations.

3.7.1 Covariance Matrix Construction

The Gaussian state of the coupled waveguide system is fully characterized by its second-order moments. These moments can be equivalently represented through the covariance matrix, which provides a compact and physically transparent description of the system's quantum correlations.

Using the analytically obtained moments $n_1(t)$, $n_2(t)$, $m_1(t)$, $m_2(t)$, $c(t)$, and $s(t)$, the covariance matrix can be written in block form as

$$\boldsymbol{\sigma}(t) = \begin{pmatrix} \mathbf{A}(t) & \mathbf{C}(t) \\ \mathbf{C}^T(t) & \mathbf{B}(t) \end{pmatrix},$$

where the 2×2 submatrices $\mathbf{A}$ and $\mathbf{B}$ describe the local fluctuations of modes a_1 and a_2 , respectively, while $\mathbf{C}$ contains the inter-mode correlations.

The local covariance blocks are given by

$$\mathbf{A}(t) = \begin{pmatrix} n_1(t) + \frac{1}{2} + m_1(t) & 0 \\ 0 & n_1(t) + \frac{1}{2} - m_1(t) \end{pmatrix}, \quad \mathbf{B}(t) = \begin{pmatrix} n_2(t) + \frac{1}{2} + m_2(t) & 0 \\ 0 & n_2(t) + \frac{1}{2} - m_2(t) \end{pmatrix}$$

The inter-mode correlation block is expressed as

$$\mathbf{C}(t) = \begin{pmatrix} s(t) + c(t) & 0 \\ 0 & s(t) - c(t) \end{pmatrix}.$$

This block-structured covariance matrix is the starting point for evaluating both Gaussian entanglement and quantum steering.

3.8 Gaussian Entanglement

3.8.1 Logarithmic Negativity

To quantify bipartite entanglement between the two waveguide modes, we employ the logarithmic negativity, which is a computable entanglement monotone for Gaussian

states. It is defined in terms of the symplectic eigenvalues of the partially transposed covariance matrix.

For a two-mode Gaussian state, the smallest symplectic eigenvalue $\tilde{\nu}_-$ of the partially transposed covariance matrix is given by

$$\tilde{\nu}_- = \sqrt{\frac{\tilde{\Delta} - \sqrt{\tilde{\Delta}^2 - 4 \det \boldsymbol{\sigma}}}{2}},$$

where

$$\tilde{\Delta} = \det \mathbf{A} + \det \mathbf{B} - 2 \det \mathbf{C}.$$

The logarithmic negativity is then defined as

$$E_N = \max [0, -\log_2 (2\tilde{\nu}_-)] .$$

A nonzero value of E_N signals the presence of Gaussian entanglement between the two waveguide modes.

3.9 Gaussian Quantum Steering

3.9.1 Steering Quantification

A quantitative measure of Gaussian steering

3.10 Squeezed States Entanglement and Steering Quantification

Chapter 4

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Chapter 5

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Chapter 6

Conclusions

Moment Equations for the Equal-Loss Two-Mode Coupler

We consider two bosonic modes with annihilation operators $\hat{a}_1$ and $\hat{a}_2$, coupled through the passive beam-splitter Hamiltonian

$$\hat{H} = J(\hat{a}_1^\dagger \hat{a}_2 + \hat{a}_2^\dagger \hat{a}_1),$$

and subject to equal damping rates

$$\kappa_1 = \kappa_2 = \kappa.$$

The relevant complex Gaussian moments are defined as

$$\begin{aligned} n_1 &= \langle \hat{a}_1^\dagger \hat{a}_1 \rangle, & n_2 &= \langle \hat{a}_2^\dagger \hat{a}_2 \rangle, \\ m_1 &= \langle \hat{a}_1 \hat{a}_1 \rangle, & m_2 &= \langle \hat{a}_2 \hat{a}_2 \rangle, \\ s &= \langle \hat{a}_1 \hat{a}_2 \rangle, & c &= \langle \hat{a}_1^\dagger \hat{a}_2 \rangle. \end{aligned}$$

Using the Heisenberg–Langevin equations (noise averages vanish for a vacuum bath), one obtains the coupled first-order differential equations

$$\begin{aligned} \dot{n}_1 &= -\kappa n_1 + iJ(c^* - c), \\ \dot{n}_2 &= -\kappa n_2 + iJ(c - c^*), \\ \dot{m}_1 &= -\kappa m_1 - 2iJ s, \\ \dot{m}_2 &= -\kappa m_2 - 2iJ s, \\ \dot{s} &= -\kappa s - iJ(m_1 + m_2), \\ \dot{c} &= -\kappa c + iJ(n_2 - n_1). \end{aligned}$$

The initial conditions used throughout this work are

$$\begin{aligned} n_1(0) &= n_{10}, & n_2(0) &= n_{20}, \\ m_1(0) &= m_{10}, & m_2(0) &= m_{20}, \\ s(0) &= 0, & c(0) &= 0, \end{aligned}$$

corresponding to uncorrelated input Gaussian states.

Analytic Solution Under Equal Loss

The system above admits a closed-form solution when $\kappa_1 = \kappa_2 = \kappa$. Define the constants

$$\begin{aligned} C_1 &= n_1(0) + \frac{1}{2}, & C_2 &= n_2(0) + \frac{1}{2}, \\ D_1 &= m_1(0), & D_2 &= m_2(0). \end{aligned}$$

Then the exact solutions for all Gaussian moments are

$$n_1(t) = \frac{e^{-\kappa t}}{2} \left[(C_1 + C_2 - 1) + (C_1 - C_2) \cos(2Jt) \right],$$

$$n_2(t) = \frac{e^{-\kappa t}}{2} \left[(C_1 + C_2 - 1) - (C_1 - C_2) \cos(2Jt) \right],$$

$$c(t) = +\frac{i e^{-\kappa t}}{2} (C_2 - C_1) \sin(2Jt),$$

$$m_1(t) = \frac{e^{-\kappa t}}{2} \left[(D_1 + D_2) \cos(2Jt) + (D_1 - D_2) \right],$$

$$m_2(t) = \frac{e^{-\kappa t}}{2} \left[(D_1 + D_2) \cos(2Jt) + (D_2 - D_1) \right],$$

$$s(t) = -\frac{i e^{-\kappa t}}{2} (D_1 + D_2) \sin(2Jt).$$

These expressions are used directly in the numerical evaluations of Gaussian steering and entanglement throughout this thesis. what is issue with this

Chapter 7

Model System and Theoretical Formulations

7.1 physical system configuration

our system consist of two identical single mode waveguides which are placed parallel to each other in a close distance such that their evanescent feild overlap and enable coupling between them. The optical modes in each waveguide are represented by creation and anihilation operator a_1 and a_2 where a_1 represents optical mode in waveguide a_1 and a_2 represents optical mode in waveguide a_2 . Each waveguide mode act as a bosonic harmonic oscillator.

7.1.1 system Hamiltonian

The hamiltonian for our system is given by

$$H = \hbar\omega \left(a_1^\dagger a_1 + a_2^\dagger a_2 \right) + \hbar J \left(a_1^\dagger a_2 + a_2^\dagger a_1 \right). \quad (7.1)$$

Here, a_1 and a_2 are the bosonic field operators for the optical modes in each waveguide, satisfying the commutation relations $[a_1, a_1^\dagger] = 1$ and $[a_2, a_2^\dagger] = 1$.

the symbol ω is angular frequency of the each waveguide mode and the parameter J in the second part of the hamiltonian represents the coupling strength between the two waveguides and its value depends on the seperation distance between the waveguides.

The hamiltonian consists of two main parts:

Free energy terms: correspond to the free energy of the individual waveguide modes,

$$\hbar\omega \left(a_1^\dagger a_1 + a_2^\dagger a_2 \right).$$

Coupling terms: account for the evanescent coupling between the two waveguide modes,

$$\hbar J \left(a_1^\dagger a_2 + a_2^\dagger a_1 \right).$$

This term is responsible for the transfer of energy between the two waveguide modes and is the main focus of interest in this work.

we can determine how this system evolve with loss by quantum louville equation.

$$\dot{\rho} = -\frac{i}{\hbar} [H, \rho] + \mathcal{L}\rho \quad (7.2)$$

this equation is known as quantum Liouville equation. It has two parts:

The Commutator:

$$[H, \rho]$$

which generates time evolution and is responsible for oscillatory exchange of energy and quantum correlation between the waveguides due to J .

Lindblad($L\rho$) term : deals with material loss, decoherence and decays due to interaction with the environment

$$\begin{aligned} \mathcal{L}\rho = & -k_1 \left(a_1^\dagger a_1 \rho + \rho a_1^\dagger a_1 - 2a_1 \rho a_1^\dagger \right) \\ & - k_2 \left(a_2^\dagger a_2 \rho + \rho a_2^\dagger a_2 - 2a_2 \rho a_2^\dagger \right), \end{aligned} \quad (7.3)$$

k_1 and k_2 represent the loss rates of modes a and b , and $a, a^\dagger$ are the annihilation and creation operators for mode a , while ρ is the density matrix. b and $b^\dagger$ are vice versa for mode b .

7.2 Dynamical equations of the model system

The time evolution of any system observable O is obtained from

$$\frac{d}{dt}\langle O \rangle = \text{Tr}(O\dot{\rho}) = -i\langle [O, H] \rangle + \langle \mathcal{L}^\dagger(O) \rangle. \quad (7.4)$$

7.2.1 Equations of motion for the moments

Using

$$\frac{d}{dt}\langle O \rangle = \text{Tr}(O\dot{\rho}),$$

Equation number(3.2), the coupled equations of motion becomes:

$$\frac{d}{dt}\langle a_1^\dagger a_1 \rangle = iJ\left(\langle a_2^\dagger a_1 \rangle - \langle a_1^\dagger a_2 \rangle\right) - k_1\langle a_1^\dagger a_1 \rangle, \quad (7.5)$$

$$\frac{d}{dt}\langle a_2^\dagger a_2 \rangle = iJ\left(\langle a_1^\dagger a_2 \rangle - \langle a_2^\dagger a_1 \rangle\right) - k_2\langle a_2^\dagger a_2 \rangle, \quad (7.6)$$

$$\frac{d}{dt}\langle a_1^\dagger a_2 \rangle = iJ\left(\langle a_2^\dagger a_2 \rangle - \langle a_1^\dagger a_1 \rangle\right) - \frac{k_1 + k_2}{2}\langle a_1^\dagger a_2 \rangle, \quad (7.7)$$

$$\frac{d}{dt}\langle a_2^\dagger a_1 \rangle = iJ\left(\langle a_1^\dagger a_1 \rangle - \langle a_2^\dagger a_2 \rangle\right) - \frac{k_1 + k_2}{2}\langle a_2^\dagger a_1 \rangle, \quad (7.8)$$

$$\frac{d}{dt}\langle a_1 a_1 \rangle = -2iJ\langle a_1 a_2 \rangle - k_1\langle a_1 a_1 \rangle, \quad (7.9)$$

$$\frac{d}{dt}\langle a_2 a_2 \rangle = -2iJ\langle a_1 a_2 \rangle - k_2\langle a_2 a_2 \rangle, \quad (7.10)$$

$$\frac{d}{dt}\langle a_1 a_2 \rangle = -iJ\left(\langle a_1 a_1 \rangle + \langle a_2 a_2 \rangle\right) - \frac{k_1 + k_2}{2}\langle a_1 a_2 \rangle. \quad (7.11)$$

so we have got seven differential equations, now we are moving towards the solutions of these equations

7.2.2 analytical solutions of the moment equations

For identical waveguides, we set

$$k_1 = k_2 = \kappa. \quad (7.12)$$

Without imposing any restriction on the initial correlations, the seven second-order moments admit the following closed-form solutions:

$$\begin{aligned}\langle a_1^\dagger a_1 \rangle(t) = & \frac{e^{-\kappa t}}{2} \left[\langle a_1^\dagger a_1 \rangle_0 + \langle a_2^\dagger a_2 \rangle_0 + (\langle a_1^\dagger a_1 \rangle_0 - \langle a_2^\dagger a_2 \rangle_0) \cos(2Jt) \right. \\ & \left. + i(\langle a_2^\dagger a_1 \rangle_0 - \langle a_1^\dagger a_2 \rangle_0) \sin(2Jt) \right],\end{aligned}\quad (7.13)$$

$$\begin{aligned}\langle a_2^\dagger a_2 \rangle(t) = & \frac{e^{-\kappa t}}{2} \left[\langle a_1^\dagger a_1 \rangle_0 + \langle a_2^\dagger a_2 \rangle_0 - (\langle a_1^\dagger a_1 \rangle_0 - \langle a_2^\dagger a_2 \rangle_0) \cos(2Jt) \right. \\ & \left. - i(\langle a_2^\dagger a_1 \rangle_0 - \langle a_1^\dagger a_2 \rangle_0) \sin(2Jt) \right],\end{aligned}\quad (7.14)$$

$$\langle a_1^\dagger a_2 \rangle(t) = e^{-\kappa t} \left[\langle a_1^\dagger a_2 \rangle_0 \cos(2Jt) - \frac{i}{2} (\langle a_1^\dagger a_1 \rangle_0 - \langle a_2^\dagger a_2 \rangle_0) \sin(2Jt) \right], \quad (7.15)$$

$$\langle a_2^\dagger a_1 \rangle(t) = e^{-\kappa t} \left[\langle a_2^\dagger a_1 \rangle_0 \cos(2Jt) + \frac{i}{2} (\langle a_1^\dagger a_1 \rangle_0 - \langle a_2^\dagger a_2 \rangle_0) \sin(2Jt) \right], \quad (7.16)$$

$$\begin{aligned}\langle a_1 a_1 \rangle(t) = & \frac{e^{-\kappa t}}{2} \left[\langle a_1 a_1 \rangle_0 - \langle a_2 a_2 \rangle_0 + (\langle a_1 a_1 \rangle_0 + \langle a_2 a_2 \rangle_0) \cos(2Jt) \right. \\ & \left. - 2i \langle a_1 a_2 \rangle_0 \sin(2Jt) \right],\end{aligned}\quad (7.17)$$

$$\begin{aligned}\langle a_2 a_2 \rangle(t) = & \frac{e^{-\kappa t}}{2} \left[-(\langle a_1 a_1 \rangle_0 - \langle a_2 a_2 \rangle_0) + (\langle a_1 a_1 \rangle_0 + \langle a_2 a_2 \rangle_0) \cos(2Jt) \right. \\ & \left. - 2i \langle a_1 a_2 \rangle_0 \sin(2Jt) \right],\end{aligned}\quad (7.18)$$

$$\langle a_1 a_2 \rangle(t) = e^{-\kappa t} \left[\langle a_1 a_2 \rangle_0 \cos(2Jt) - \frac{i}{2} (\langle a_1 a_1 \rangle_0 + \langle a_2 a_2 \rangle_0) \sin(2Jt) \right]. \quad (7.19)$$

Although seven coupled moment equations are obtained, they do not all correspond to independent dynamical variables. In particular, the inter-mode coherence terms $\langle a_1^\dagger a_2 \rangle$ and $\langle a_2^\dagger a_1 \rangle$ are related by Hermitian conjugation, i.e., $\langle a_2^\dagger a_1 \rangle = \langle a_1^\dagger a_2 \rangle^*$. Therefore, only six independent moments are sufficient to fully describe the system dynamics, while the remaining coherence follows directly from this relation.

Define the constants

$$C_1 = \langle a_1^\dagger a_1 \rangle_0 + \frac{1}{2}, \quad C_2 = \langle a_2^\dagger a_2 \rangle_0 + \frac{1}{2},$$

$$D_1 = \langle a_1 a_1 \rangle_0, \quad D_2 = \langle a_2 a_2 \rangle_0.$$

Then the exact solutions for all Gaussian moments are

$$n_1(t) = \frac{e^{-\kappa t}}{2} \left[(C_1 + C_2 - 1) + (C_1 - C_2) \cos(2Jt) \right],$$

$$n_2(t) = \frac{e^{-\kappa t}}{2} \left[(C_1 + C_2 - 1) - (C_1 - C_2) \cos(2Jt) \right],$$

$$c(t) = +\frac{i e^{-\kappa t}}{2} (C_2 - C_1) \sin(2Jt),$$

$$m_1(t) = \frac{e^{-\kappa t}}{2} \left[(D_1 + D_2) \cos(2Jt) + (D_1 - D_2) \right],$$

$$m_2(t) = \frac{e^{-\kappa t}}{2} \left[(D_1 + D_2) \cos(2Jt) + (D_2 - D_1) \right],$$

$$s(t) = -\frac{i e^{-\kappa t}}{2} (D_1 + D_2) \sin(2Jt).$$

where, For convenience, we introduce the following time-dependent second-order moments:

$$n_1(t) = \langle a_1^\dagger a_1 \rangle(t), \quad n_2(t) = \langle a_2^\dagger a_2 \rangle(t),$$

$$c(t) = \langle a_1^\dagger a_2 \rangle(t), \quad m_1(t) = \langle a_1 a_1 \rangle(t), \quad m_2(t) = \langle a_2 a_2 \rangle(t), \quad s(t) = \langle a_1 a_2 \rangle(t).$$

These expressions are used directly in the numerical evaluations of Gaussian steering and entanglement throughout this thesis. what is issue with this

For independently prepared input modes (no initial inter-mode correlations), $\langle a_1^\dagger a_2 \rangle_0 = \langle a_1 a_2 \rangle_0 = 0$, and the corresponding terms drop out, yielding the simplified expressions used later in this thesis. ‘

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